Polynomials with Predicates: a Dialectica reading*

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Abstract

A Dialectica object is a set of witnesses, a set of challenges and a relation saying which witness withstands which challenge, with morphisms running forward on the witnesses and backward on the challenges. Forgetting the relation leaves a polynomial functor of a special kind, one with a set of directions attached to each position. Dialectica Categories (1991) give two constructions of this shape, differing over whether the backward map may see the witness or not. Under this reading the Dialectica connectives, three tensors, two implications, and the additives, are operations that the theory of polynomial functors has since renamed, with the exception of par the multiplicative disjunction. Further work on Dialectica uncovered 'sequential' connectives, inspired by linear models of state, that also have polynomial shapes. We can easily put a predicate on the substitution product of polynomials, and this gives a new Dialectica sequencing connective in which the witness of the second component is chosen in the light of the challenge to the first. The two tensors are then related by a single map, the middle interchange, and everything else follows from it: a context may be given more order than it has, the sequential modality is determined rather than postulated, and whatever may be reused in parallel may be reused in sequence. This is how combining things in parallel and combining them in order are made to fit together. And, if you like the label, this is what a normal duoidal category buys. The Dialectica construction carries the structure.

*This note grew out of the 2022 Mathematics Research Community on applied category theory, where one of the working groups looked at the relationship between the Dialectica construction and polynomials. That group's own account of it is Dorta, Jarvis and Niu [8].

1 Dialectica constructions

The Dialectica construction reads a proposition as a two-move dialogue. The proposition is presented by a set U of witnesses, a set X of counterexamples or challenges, and a relation α saying which witness withstands which challenge, so that the proposition asserts $\exists u \in U.\, \forall x \in X.\, \alpha(u, x)$: some witness withstands every challenge to it. To pass from one proposition to another is to produce a witness for the second from a witness for the first, and to turn every challenge that might be raised against what was produced back into a challenge to what one started with. So the data of a morphism is a pair of maps running in opposite directions, one forward on witnesses and one backward on counterexamples, together with a condition saying that the backward map turns challenges back well.

Definition 1 (**Dial**($\mathcal{C}$), 1991)**.** Let $\mathcal{C}$ be a cartesian closed category with finite products. An object of the category **Dial**($\mathcal{C}$) is a triple $A = (U, X, \alpha)$ with U and X objects of $\mathcal{C}$ and α a subobject of $U \times X$, read as saying which witness $u \in U$ withstands which counterexample $x \in X$; we say counterexample and challenge interchangeably. A morphism of **Dial**($\mathcal{C}$) from A to $B = (V, Y, \beta)$ is a pair

$$f \colon U \longrightarrow V, \qquad F \colon U \times Y \longrightarrow X$$

in $\mathcal{C}$ such that $\langle \pi_1, F \rangle^* \alpha \leq (f \times 1_Y)^* \beta$ as subobjects of $U \times Y$. In $\mathcal{C} = \mathbf{Set}$ this says that $\alpha(u, F(u, y))$ implies $\beta(fu, y)$: if u withstands the challenge $F(u, y)$ that y was turned into, then fu withstands y. Composition is $(g, G) \circ (f, F) = (gf,\ (u, z) \mapsto F(u, G(fu, z)))$.[1]

The comparison below is with a construction on sets, so $\mathcal{C}$ is **Set** from here on and α is an ordinary relation. The Dialectica construction comes in two forms, both introduced in [10]. They differ over what the backward map is allowed to see. In the form just defined, F takes the witness u as well as the challenge y, so the challenge it hands back may be chosen differently for different witnesses; this is the Gödel Dialectica[4]. In the other form F is a single function $Y \to X$, fixed once and for all, and this is the Girard Dialectica [9], written **Gir**(**Set**) here.

The shape of that data, forward on one component and backward on the other, is not peculiar to the Dialectica construction, and it is worth looking at where else it occurs before returning to the predicates.

[1] The hypothesis is stronger than the category needs: finite products are enough to define **Dial**($\mathcal{C}$), and the cartesian closure is used for the structure carried by it in this note.

2 Polynomials and monomials

A polynomial functor is a set together with a set at each of its elements. The elements are called *positions* and the sets attached to them are the sets of *directions* available at each position, so the datum is a set I and a family $(X_i)_{i\in I}$: the same datum as a container in [3], or as an interface at which one may be in any of the positions i and may then choose among the directions in X_i. It is a functor because it acts on a set S by $S \mapsto \sum_{i\in I} S^{X_i}$, but nothing below uses that action; positions and directions are what we need. The category they form is called **Poly** [11].

A morphism of polynomials goes forward on positions and backward on directions. Given $(I, (X_i))$ and $(J, (Y_j))$, it consists of a function $\varphi_1 \colon I \to J$ and, for each position i, a function $\varphi_i^\sharp \colon Y_{\varphi_1 i} \to X_i$: having sent the position i to the position $\varphi_1 i$, it must send each direction available there back to a direction available at i. The backward function is indexed by the position, so it may differ from one position to another.

The notation is the reason for the name. Write y^X for the polynomial with a single position carrying the directions X, and add such terms to put positions side by side, so that

$$p \;=\; \sum_{i\in I} y^{X_i}$$

is the polynomial above; a polynomial with three positions, two of them carrying three directions and one carrying one, is $2y^3 + y$. We write $p(1)$ for its set of positions and $p[i]$ for the directions at i.

Given this terminology *monomials* are where the comparison with the Dialectica construction lives, and the reason is the quantifier. A Dialectica object asserts $\exists u \in U.\, \forall x \in X.\, \alpha(u, x)$, and the $\forall$ ranges over the same X whatever witness the $\exists$ produces: the challenges available to the opponent are fixed in advance, and only the predicate tells one witness from another. A general polynomial lets the second set vary with the first, $\exists i \in I.\, \forall x \in X_i$, which is a proposition of a different shape. So the monomials are the polynomials that present propositions in the Dialectica sense, and an arbitrary polynomial is what one gets by letting the challenges depend on the witness.

Definition 2 (monomial)**.** A polynomial is a *monomial* when the directions do not vary with the position: $p = Uy^X$, with $p(1) = U$ and $p[u] = X$ for every u. We write $\mathbf{Poly}_{\mathsf{mono}}$ for the full subcategory of monomials.

Writing the same set X at every position is a matter of how the polynomial is presented. What matters is the weaker condition that the sets at

the positions are all isomorphic, since a bijection at each position gives an isomorphism $y^X \cong y^{X'}$ whenever $X \cong X'$. So a polynomial whose direction sets agree up to isomorphism is already a monomial, and fixing one set X is a choice of notation, exactly as fixing the set of counterexamples of a Dialectica object is, and $\mathbf{Poly}_{\mathsf{mono}}$ is closed under isomorphism in the sense that matters.

Monomials are the more fundamental of the two notions. The representables y^X are the building blocks; a monomial is a sum of copies of a single block, and a general polynomial is what one gets by setting blocks of different sizes side by side. So the logic lives on the building blocks: a Dialectica object is a sum of copies of one block carrying a predicate, and the connectives below never leave that class.

3 Dialectica and monomials

A morphism of Dialectica objects is a forward map on witnesses and a backward map on challenges that may see the witness, subject to the condition on the predicates. Drop that condition and what is left is a function $f\colon U \to V$ together with a U-indexed family of functions $Y \to X$, which is a single function $F\colon U \times Y \to X$: precisely a morphism of monomials $Uy^X \to Vy^Y$. Call the monomial Uy^X the *shape* of the Dialectica object (U, X, α): what is left of it when the predicate is forgotten. Write $|-|\colon \mathbf{Dial}(\mathbf{Set}) \to \mathbf{Poly}_{\mathsf{mono}}$ for the functor taking each object to its shape and each morphism to its underlying pair, which is faithful.

The two theorems below are what the rest of this note rests on. Their proofs are a matter of reading definitions, but what they say is that every connective of the two constructions splits into a polynomial operation on the shapes and an operation on the predicates, and that the shapes stay inside the monomials throughout. Once the two parts are separated they can be varied independently: the shape can be changed while the predicate is kept, or the shape kept while the predicate takes its values somewhere other than in the boolean. So the constructions stop being a list of definitions to be checked one at a time and become a landscape of variations to be surveyed, which is what the sections below do.

Theorem 3 (connectives of the Gödel Dialectica)**.** *Let $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ be objects of $\mathbf{Dial}(\mathbf{Set})$. The connectives of $\mathbf{Dial}(\mathcal{C})$ (from [10]) read as follows in the canonical case $\mathcal{C} = \mathbf{Set}$, and the shape of each is a monomial.*

1. *The Dialectica tensor $A \otimes B$ has witnesses $U \times V$ and challenges $X \times Y$: a witness is a witness of each component and a challenge is a challenge to each, so $(U \times V,\ X \times Y,\ \wedge)$. Its unit is $I = (1, 1, e)$, where e is the unit of the lineale, of shape y. The shape $(U \times V)y^{X \times Y}$ is what* **Poly** *calls the* Dirichlet product*.*

2. *The Dialectica linear implication $A \multimap B$ has for witnesses the pairs of maps $(f\colon U \to V,\ F\colon U \times Y \to X)$, which is the data of a morphism from A to B with the condition on predicates not yet imposed, and for challenges the pairs (u, y) of a witness of A and a challenge to B, so $(V^U \times X^{U \times Y},\ U \times Y,\ \beta^\alpha)$. A witness (f, F) withstands (u, y) when u withstanding $F(u, y)$ implies fu withstanding y, so the witnesses that withstand every challenge are exactly the morphisms $A \to B$. It is the closure of the original dialectica tensor:*

$$\mathbf{Dial}(\mathbf{Set})(C \otimes A,\ B) \ \cong\ \mathbf{Dial}(\mathbf{Set})(C,\ A \multimap B).$$

   Its shape $\left(V^U \times X^{U \times Y}\right) y^{U \times Y}$ is the closure that **Poly** *has for the same product, where the corresponding adjunction holds for every polynomial C, monomial or not.*

3. *The Dialectica additive conjunction $A \,\&\, B$ has witnesses $U \times V$ and challenges $X + Y$, so $(U \times V,\ X + Y,$ choose$)$: a challenge picks one component and challenges it. Its unit is $(1, 0)$, of shape y^0. The shape $(U \times V)y^{X+Y}$ is the cartesian product in* $\mathbf{Dial}(\mathbf{Set})$ *and in* **Poly***.*

4. *The additive disjunction $A \oplus B$ has witnesses $U + V$ and challenges $X \times Y$, a witness of one component being challenged by the corresponding component of the pair. It receives both injections, but it is a weak coproduct only in* $\mathbf{Dial}(\mathbf{Set})$*, while it is a coproduct in* $\mathbf{Gir}(\mathbf{Set})$*: there the copairing of (f, F) and (g, G) has backward part $\langle F, G\rangle$, which is available because a backward map is a single function, whereas here a backward map out of $A \oplus B$ is defined on $(U + V) \times Z$ and has nothing to supply its Y component at a witness from U.*

Proof. Items (1), (3) and (4) are the connectives of [10] restated, and in each the witnesses, the challenges and the predicate are as displayed. For (2), let $C = (W, Z, \gamma)$ be any object. A morphism $C \otimes A \to B$ is a map f on witnesses and a map F on challenges,

$$f\colon W \times U \to V, \qquad F\colon (W \times U) \times Y \to Z \times X,$$

and, writing $F = (F_1, F_2)$, its condition is

$$\gamma\big(w, F_1(w,u,y)\big) \otimes \alpha\big(u, F_2(w,u,y)\big) \;\sqsubseteq\; \beta\big(f(w,u), y\big).$$

A morphism $C \to A \multimap B$ is a map on witnesses sending w to a pair $(h_1(w), h_2(w))$ with $h_1(w)\colon U \to V$ and $h_2(w)\colon U \times Y \to X$, together with a map $H\colon W \times (U \times Y) \to Z$ on challenges, and its condition is

$$\gamma\big(w, H(w,u,y)\big) \;\sqsubseteq\; \Big[\, \alpha\big(u, h_2(w)(u,y)\big) \multimap \beta\big(h_1(w)(u), y\big) \,\Big].$$

Transposing in W matches the two sets of data,

$$h_1(w)(u) = f(w,u), \qquad h_2(w)(u,y) = F_2(w,u,y),$$

$$H(w,u,y) = F_1(w,u,y),$$

and matches the two conditions by residuation in the lineale, since $a \otimes b \sqsubseteq c$ holds exactly when $a \sqsubseteq (b \multimap c)$. Both correspondences are natural in C. Each shape is a monomial because the challenges of the object just constructed do not vary with its witness. □

The second construction, where the backward map is a single function $Y \to X$, has connectives of its own, and the same holds of them.

Theorem 4 (connectives of the Girard Dialectica)**.** *Let A and B be dialectica objects as above, now taken in* **Gir**(**Set**).

1. *The cross product $A \otimes_{\mathbf{Gir}} B$ has witnesses $U \times V$ and for challenges the pairs of functions $(f\colon V \to X,\ g\colon U \to Y)$, each choosing a challenge to one component in the light of the witness of the other; a pair (u,v) withstands (f,g) when u withstands fv and v withstands gu, so $(U \times V,\ X^V \times Y^U,\ \text{cross})$. Its unit is I again. Its shape is $(U \times V)\, y^{X^V \times Y^U}$.*

2. *The cross product is closed, the implication having witnesses the pairs $(f\colon U \to V,\ F\colon Y \to X)$ and challenges $U \times Y$, with (f,F) withstanding (u,y) when u withstanding Fy implies fu withstanding y, so $(V^U \times X^Y,\ U \times Y,\ \beta^\alpha)$. Its shape is $(V^U \times X^Y)\, y^{U \times Y}$; the adjunction is taken with respect to the morphisms of this construction, whose backward maps do not see the witness.*

3. *The coproduct $A \oplus B$ has witnesses $U + V$ and challenges $X \times Y$, so $(U+V,\ X \times Y,\ \text{choose})$, of shape $(U+V)\, y^{X \times Y}$. Here it is a genuine coproduct, for the reason given in Theorem 3(4).*

4. *The par $A \parr B$ has for witnesses the pairs of functions $(f\colon Y \to U,\ g\colon X \to V)$ and for challenges the pairs (x, y), with (f, g) withstanding (x, y) when fy withstands x or gx withstands y, so $(U^Y \times V^X,\ X \times Y,\ \vee)$. Its shape is $(U^Y \times V^X)\, y^{X \times Y}$, which puts function spaces among the positions and a plain product among the directions; we have not found that combination among the monoidal structures listed in [12].*

Proof. Each item is the corresponding connective of the second construction of [10], whose backward maps are single functions rather than families indexed by the witness; the witnesses, challenges and predicate are as displayed, and in each case the challenges again do not vary with the witness. □

The polynomial connectives on monomials are the Dialectica ones, and they were written down thirty years earlier. The predicate is not something the items above add to them: it was there from the start, and it is what the functor $|-|$ forgets. Each connective computes it alongside the shape, $\alpha \wedge \beta$ pointwise for the tensor, $\alpha(u, F(u, y)) \Rightarrow \beta(fu, y)$ for the implication, and α on the left summand with β on the right for the conjunction, so that $|-|$ carries each connective to the corresponding polynomial structure and the adjunction $\otimes \dashv \multimap$ in $\mathbf{Dial}(\mathbf{Set})$ is the adjunction of item (2) together with the residuation $\gamma \wedge \alpha \leq \beta$ iff $\gamma \leq (\alpha \Rightarrow \beta)$ applied pointwise.

The shape of the cross product (the tensor product of $\mathbf{Gir}(\mathbf{Set})$) appears in **Poly** without a name, among the monoidal products collected in [12].

3.1 More tensors

The Dialectica objects allow many more monoidal bifunctors. Blass worked with the objects of $\mathbf{Gir}(\mathbf{Set})$, which he calls $\mathcal{PV}$, and defined on them a further tensor product [1]. He arrives at it from the set-theoretic reading of the morphisms, where a proof of an inequality between three cardinal characteristics is a morphism out of a composite of two objects, and the composite asks a first question, receives an answer, and chooses the second question in the light of that answer.

Definition 5 (Blass)**.** Given A and B as before, let

$$A \rtimes B \;=\; \big(U \times V,\ X^V \times Y,\ \alpha(u, fv) \wedge \beta(v, y)\big),$$

a witness being a pair (u, v) and a challenge a pair (f, y) with $f\colon V \to X$, which withstand each other when u withstands fv and v withstands y.

Blass runs the dependence the other way: for him the second question is chosen in the light of the answer to the first, so his challenges are $X \times Y^U$. The two are one operation with the roles of the components interchanged, and the form above, where the challenge to the first component is the one chosen, is the one used below.

The witnesses are a plain product, but the challenges have one exponential, X^V in place of X. The same operation is one of the two further monoidal products listed at the end of [8]. On homogeneous polynomials over the booleans, the one with function spaces on both sides is the cross product of Theorem 4(1), the multiplicative conjunction of the Girard Dialectica, where it interprets $\otimes$ and stands with the par in a de Morgan pair [9, 10]; the one with a function space on a single side is Blass's, interpreting a reduction that asks a question, reads the answer, and asks a second in the light of it [1].[2]

The definition of Blass' tensor has a mirror image, and this is the connective that **Poly** has and Dialectica never considered before, although it knew the connective was there. What told it so is a map it already had. The two tensors of the thesis are related by

$$A \otimes B \longrightarrow A \otimes_{\mathbf{Gir}} B,$$

the identity on witnesses, sending a challenge (f, g) of the cross product back to the challenge (fv, gu) at the witness (u, v): the constructions already contained a passage from no dependence at all to dependence on both sides, in the challenge coordinate, and Blass's tensor is the step halfway along it. Linear negation carries the cross product to the par, whose witnesses are $U^Y \times V^X$, so the same passage exists in the witness coordinate, from $\otimes$ to $\parr$. Its halfway step was never written down, and what was missing was not the shape but a reading for it: a shape becomes a connective when something says what it means, and what says so for this one is the calculus of Section 6, whose ordered contexts fix what "before" has to satisfy. Instead of letting a challenge be chosen in the light of the other component's witness, let a witness be chosen in the light of the other component's challenge.

Proposition 6. *Given A and B dialectica objects, consider the bifunctor*

$$A \triangleleft B \;=\; \big(U \times V^X,\; X \times Y,\; \alpha \triangleleft \beta\big), \tag{1}$$

[2] Dorta, Jarvis and Niu ask whether interpretations can be found for these two products [8]. Both have had them for decades: the two-sided one has been the multiplicative conjunction of a model of classical linear logic since 1989 [9, 10], and the one-sided one is Blass's sequential composition [1]. The same paper identifies the homogeneous polynomials over the booleans with **Dial**(**Set**), citing [10].

whose witnesses are a witness of A together with a choice of witness of B for each challenge to A, and whose challenges are a challenge x to the first component paired with a challenge y to the witness that x elicited. It is defined on the objects of **Gir**(**Set**)*, as $\rtimes$ is, and equally on those of* **Dial**(**Set**)*, the two constructions having the same objects. On shapes it is the* substitution product *of* **Poly**.

Proof. A witness of the composite is a witness u of A together with a rule φ choosing, for each challenge x that may be raised against u, a witness of B to be confronted next; so the witnesses are $U \times V^X$. A challenge to it names the challenge x to be raised first and a challenge y to whatever x elicited, and since the challenges to B do not vary with its witness that second choice is made from Y, so the challenges are $X \times Y$. Given $(f, F)\colon A \to A'$ and $(g, G)\colon B \to B'$, the witness (u, φ) is carried to $\big(fu,\ x' \mapsto g\varphi F(u, x')\big)$ and the challenge (x', y') back to $\big(F(u, x'),\ G(\varphi F(u, x'), y')\big)$. These formulas are functorial in each argument. For **Gir**(**Set**), where the backward maps are functions $(F' \to X)$ and $(G' \to Y)$ independent of the witnesses, the same construction restricts to

$$(u, \phi) \longmapsto \big(fu,\ x' \mapsto g\phi(Fx')\big), \qquad (x', y') \longmapsto (Fx', Gy'),$$

so the backward map is again independent of the witness. The condition on the predicates is checked in Section 4, where the relation $\alpha \triangleleft \beta$ is given. Forgetting the predicates, the witnesses and challenges just computed are the positions and directions of the substitution product of the two monomials. □

The relation $\alpha \triangleleft \beta$ is written out in Section 4, where the values are allowed to lie in a lineale, since the two predicates are combined there in a fixed order. Table 1 collects the connectives of the two constructions together with the two of this section, and names the polynomial operation carried by each shape.

Not every operation of **Poly** appears in the table. Among those omitted are the free monad, the derivative and the coproduct of **Poly**, which do not in general preserve monomials. The two Dialectica constructions also have their own additive disjunctions, as recorded in the table.

The Dialectica reading is not a proposal to work with polynomials with predicates. The predicate is one dial and the shape is another, and the shape has settings this note leaves alone. The challenges may be fixed, which is the Dialectica setting and the one used throughout; they may vary with the witness, which is the general polynomial; the witnesses may vary with

	witnesses	challenges	**Poly**
in **Dial**(**Set**)			
tensor $A \otimes B$	$U \times V$	$X \times Y$	Dirichlet product
implication $A \multimap B$	$V^U \times X^{U \times Y}$	$U \times Y$	its closure
conjunction $A \,\&\, B$	$U \times V$	$X + Y$	cartesian product
(weak) disjunction $A \oplus B$	$U + V$	$X \times Y$	—
in **Gir**(**Set**)			
tensor $A \otimes_{\mathbf{Gir}} B$	$U \times V$	$X^V \times Y^U$	unnamed in [12]
implication $A \multimap_{\mathbf{Gir}} B$	$V^U \times X^Y$	$U \times Y$	—
conjunction $A \,\&\, B$	$U \times V$	$X + Y$	cartesian product
disjunction $A \oplus B$	$U + V$	$X \times Y$	—
par $A \parr B$	$U^Y \times V^X$	$X \times Y$	—
further connectives			
Blass's $A \rtimes B$	$U \times V$	$X^V \times Y$	unnamed in [12]
substitution $A \triangleleft B$	$U \times V^X$	$X \times Y$	substitution product

Table 1: The connectives of the two Dialectica constructions, with Blass's tensor and the substitution tensor, given by their witnesses and challenges and by the polynomial operation on their shapes. A dash means we have not found the operation in the monoidal polynomial list in [12].

the challenge, which is the same dependence read in the other direction; or neither may be fixed, which asks for two indices rather than one. Which operations work in each setting is what a comparison is for, and only the first setting is treated below.

4 Substitution tensor

From here on fix a lineale L and write $\mathbf{Dial}_L(\mathbf{Set})$ for the category of Definition 1 with $\alpha\colon U \times X \to L$ and the morphism condition $\alpha(u, F(u,y)) \sqsubseteq \beta(fu, y)$; Theorem 3 is about shapes and is unaffected, and $\mathbf{Dial}(\mathbf{Set})$ is the case $L = 2$. The lineale is what the predicate of the substitution tensor needs, since two predicates are combined there in a fixed order.

In this section we want to explain the substitution tensor in Dialectica terms, so Equation (1) gives the shape and here is the predicate. A witness of $A \triangleleft B$ is a witness u of A together with a function φ assigning a witness of B to each challenge to A; a challenge to it is a pair (x, y), a challenge x to u together with a challenge y to the witness φx that x elicited. So $A \triangleleft B$ is A and then B, with the second component's witness chosen in the light of how the first was attacked, and with the two challenges made in that order.

Definition 7 ($\triangleleft$ tensor)**.** For dialectica objects $A = (U, X, \alpha)$ and $B =$

(V, Y, β) set

$$A \triangleleft B \;:=\; \big(U \times V^X,\; X \times Y,\; \alpha \triangleleft \beta\big),$$
$$(\alpha \triangleleft \beta)\big((u,\varphi),(x,y)\big) \;:=\; \alpha(u,x) \otimes \beta(\varphi x, y),$$

the tensor being that of the lineale L. Its nullary tensor is

$$I = (1, 1, e),$$

where e is the monoidal unit of the lineale L.This is the Dialectica version of the *substitution tensor* of polynomials.

The device of putting a function space into one of the two coordinates of a Dialectica object is Blass's, and Definition 5 puts it on the challenges. What is taken up here is the other coordinate, together with the algebra that comes with it: a function space among the witnesses makes the operation on shapes the substitution product of polynomials, so that associativity, the unit and the interchange with $\otimes$ are properties of that operation rather than calculations.

This is the composition product of [8] at $\mathcal{C} = 2$, where the predicate is allowed to take its values in any monoidal category and the predicate of a composite is the monoidal product of the two; the shape is theirs and so is the predicate on it, and what Section 6 adds is the reading. No exponential is asked of that category: the positions and the directions are sets throughout, and V^X is the X-indexed product of copies of V taken among them. Over a base other than **Set** it is a real demand, and it is one of the uses that the cartesian closure of Definition 1 is put to.

Proposition 8. *The connective $\triangleleft$, together with its nullary tensor I, defines a monoidal structure on $\mathbf{Dial}_L(\mathbf{Set})$, and $|A \triangleleft B| = |A| \triangleleft |B|$. It is not symmetric.*

Proof. Given $(f, F)\colon A \to A'$ and $(g, G)\colon B \to B'$, the morphism $(f, F) \triangleleft (g, G)$ acts by

$$(u,\varphi) \longmapsto \big(fu,\; x' \mapsto g\varphi F(u,x')\big),$$
$$(x',y') \longmapsto \big(F(u,x'),\; G(\varphi F(u,x'),y')\big),$$

the first components being substituted one into the other as in **Poly**. Writing $v := \varphi F(u,x')$, the required inequality is

$$\alpha(u, F(u,x')) \otimes \beta(v, G(v,y')) \;\sqsubseteq\; \alpha'(fu,x') \otimes \beta'(gv,y'),$$

which follows from the two morphism conditions and monotonicity of $\otimes$ in L; that identities and composites are preserved is immediate from the displayed formulas.

For associativity, a witness of $(A \triangleleft B) \triangleleft C$ is a witness (u, φ) of $A \triangleleft B$ together with a rule assigning a witness of C to each challenge (x, y) raised against it, and a witness of $A \triangleleft (B \triangleleft C)$ is a witness u together with a rule assigning to each challenge x a witness of $B \triangleleft C$. Currying in the second challenge matches the two, and it matches the challenges $(x, y), z$ with $x, (y, z)$; the two predicates are then $\big(\alpha(u, x) \otimes \beta(\varphi x, y)\big) \otimes \gamma(\psi(x, y), z)$ and $\alpha(u, x) \otimes \big(\beta(\varphi x, y) \otimes \gamma(\psi(x, y), z)\big)$, equal by associativity in L. For the unit, a witness of $I \triangleleft A$ is a rule assigning a witness of A to the single challenge to I, which is a witness of A, and its predicate is $e \otimes \alpha(v, y) = \alpha(v, y)$; a witness of $A \triangleleft I$ is a witness of A together with a rule with nothing to choose, and its predicate is $\alpha(u, x) \otimes e$. Non-commutativity is visible in the witnesses: $A \triangleleft B$ has $U \times V^X$ and $B \triangleleft A$ has $V \times U^Y$. □

4.1 Relating Tensors

With every connective now written as a Dialectica object, the tensors fall into two chains that leave from the same place. Start from the Dialectica tensor $\otimes$, where neither component sees the other. Let a challenge to one component be chosen in the light of the witness of the other, on one side only, and the challenges become $X^V \times Y$, which is Blass's $\rtimes$ of Definition 5; let it happen on both sides and they become $X^V \times Y^U$, which is the cross product of Theorem 4(1). Along this chain the witnesses stay a plain product. Now move the exponentials to the other coordinate instead. Let the witness of one component be chosen in the light of the challenge to the other, on one side only, and the witnesses become $U \times V^X$, which is the substitution tensor of Equation (1); on both sides they become $U^Y \times V^X$, which is the par of Theorem 4(4). Along this chain the challenges stay $X \times Y$. The shapes are those of Table 1, and the steps are morphisms.

Proposition 9. *Two chains of tensors live naturally in the two Dialectica categories.*

1. *In* **Dial**(**Set**) *there are morphisms, natural in A and B,*

$$A \otimes B \longrightarrow A \rtimes B \longrightarrow A \otimes_{\mathrm{Gir}} B,$$

which are the identity on witnesses.

2. *There are morphisms, natural in A and B,*

$$A \otimes B \longrightarrow A \triangleleft B \longrightarrow A \parr B,$$

which are the identity on challenges. The first of them is a morphism of both **Dial**(**Set**) *and* **Gir**(**Set**)*, and both $\otimes$ and $\triangleleft$ are bifunctors on either category, so that step of the chain is natural in both. The chain as a whole is a chain in* **Gir**(**Set**)*, because the par is a connective of the Girard construction and is not a bifunctor on* **Dial**(**Set**)*.*

Proof. For the first chain, send the challenge (f, y) of $A \rtimes B$ back to the challenge (fv, y) of $A \otimes B$ at the witness (u, v), and the challenge (f, g) of $A \otimes_{\mathbf{Gir}} B$ back to the challenge (f, gu) of $A \rtimes B$. These maps depend on the witness and hence are morphisms of **Dial**(**Set**). In each case the predicates agree after substitution, so the Dialectica condition holds with equality.

For the second chain, send the witness (u, v) to (u, const_v) and the witness (u, φ) to $(\mathrm{const}_u, \varphi)$, the challenges going back unchanged. Thus the backward maps are independent of the witnesses, so both are morphisms of **Gir**(**Set**) and the first is a morphism of **Dial**(**Set**) as well. For the first map the predicates agree; for the second the required condition is

$$\alpha(u, x) \wedge \beta(\phi x, y) \implies \alpha(u, x) \vee \beta(\phi x, y).$$

Naturality follows directly from the definitions of the bifunctors and the displayed maps. For the par this is naturality in **Gir**(**Set**): a par of Dialectica morphisms would have to produce a witness in $U'^{Y'}$ from one in U^Y, and the only passage from Y' to Y is the backward map of B, which asks for a witness of V that nothing here supplies. □

The distinction between the two chains is thus also a distinction between the two Dialectica categories. The comparison maps in the first chain use witness-dependent backward maps, while those in the second have witness-independent backward maps. This is why the two chains live naturally in the two different Dialectica categories. The first step of the second chain is the map called ser in Section 6.

At the level of objects, linear negation exchanges the two coordinates of an object, so it exchanges the two chains term by term: the cross product is dual to the par, Blass's $\rtimes$ is dual to the substitution tensor $\triangleleft$, and the Dialectica tensor, at the head of both, is carried to itself. The additives are monoidal structures too, but do not take part in either chain.

5 Modalities and Comonoids

The connectives so far are the multiplicatives and the additives, and the constructions carry two more. A modality says which formulas may be used again, and in a categorical model it is a comonad whose values carry comonoid structure for a tensor, the comultiplication being the map that splits a formula into two copies. There is one modality for each of two of the tensors, $\otimes$ and $\triangleleft$, and the difference between them is a monoid. Reuse in parallel is unordered, and the comonoid interpreting it is required to be commutative, so the challenges of $!A$ form the free commutative monoid on X, which is the finite multisets on it. Reuse one after another is ordered, and no commutativity is asked of the comonoid, so the challenges of $\dagger A$ form the free monoid, which is the words on X. The earlier models make the same match from the other side: with a commutative connective for "before" the modality took the free commutative monoid, and the version with a non-commutative one took plain sequences [6, 7].

Definition 10 (modalities)**.** For $A = (U, X, \alpha)$ put

$$!A := \big(U,\ M_{\mathrm{fin}}X,\ !\alpha\big), \qquad \dagger A := \big(U,\ X^*,\ \dagger\alpha\big),$$

where $M_{\mathrm{fin}}X$ is the free commutative monoid on X, that is the finite multisets on it, and X^* is the free monoid, and where $!\alpha(u, m)$ and $\dagger\alpha(u, w)$ are the $\otimes$-product in L of the values $\alpha(u, x)$ over the entries of m and of w respectively. Both act on maps by the functor applied to the backward component.

Neither modality is new, and neither is the structure they carry. Definition 10 is the ! of the thesis [10] together with the † that came with the sequencing connective of the models of state [6, 7]. What was established there is more than that they are comonads: each is a *linear exponential comonad*, a monoidal comonad whose values carry comonoid structure compatibly with the comonad and with the tensor, which is exactly what a model of a linear modality requires. The counit is given by the singleton and the comultiplication by the multiplication of the monad on the challenge side, in each case the identity on witnesses.

What is wanted here is not the comonad structure but the comonoid structure each of them carries, since that is what a modality must have if it is to interpret reuse: a formula that may be used twice has to come with a map splitting it in two, and that map has to be a comonoid comultiplication for the tensor by which the two copies are combined. Each of the two modalities has one, for a different tensor.

Proposition 11. *Let A be an object of* $\mathbf{Dial}_L(\mathbf{Set})$*. The following are comonoid structures in that category, for $\otimes$ and for $\triangleleft$ respectively.*

1. *$!A$ is a comonoid for $\otimes$, with comultiplication the diagonal on witnesses and the sum of multisets backwards, and counit the empty multiset;*
2. *$\dagger A$ is a comonoid for $\triangleleft$, with comultiplication sending u to (u, const_u) and concatenating words backwards, and counit the empty word;*
3. *the abelianisation $X^* \to M_{\mathrm{fin}}X$ is the backward part of a comonad morphism $\kappa \colon ! \to \dagger$, the identity on witnesses.*

Proof. In each case the predicate condition is an equality rather than an inequality, because the predicate of $!A$ at a multiset and of $\dagger A$ at a word is the $\otimes$-product over the entries, and concatenation, multiset sum and abelianisation all preserve that product. The comonoid laws are then laws of the monoid of challenges. Coassociativity holds because concatenating three words, and adding three multisets, may be bracketed either way; the counit laws hold because the empty word and the empty multiset are neutral, the witness component being the identity or the diagonal in every case. For (3), the abelianisation commutes with concatenation and carries the empty word to the empty multiset, which is what makes it a map of comonoids, and it commutes with the counits and comultiplications of the two comonads for the same reason. □

The proposition is not new either. Item (1) is in the thesis [10], where ! is shown to be a linear exponential comonad on the Dialectica category, and items (2) and (3) are Proposition 2 of [7], with the sequencing connective $\odot$ of that paper in place of $\triangleleft$. What is new here is only the change of tensor: the witnesses of $A \triangleleft B$ are not a plain product, so the comultiplication of $\dagger A$ has a forward part to supply as well as a backward one, and it is the constant map $u \mapsto \mathrm{const}_u$ that supplies it.

The comonoids for $\triangleleft$ deserve a paragraph of their own, because they have a reading that the comonoids for $\otimes$ do not. A comonoid for the substitution product in **Poly** is precisely a small category, by a result of Ahman and Uustalu [5]: the positions are the objects, the directions at a position are the morphisms out of that object, the counit picks the identity at each object, and the comultiplication says where each morphism lands and how two of them compose. The identification is not special to **Poly**. A comonoid for $\triangleleft$ in $\mathbf{Dial}_L(\mathbf{Set})$ is a small category enriched in L^{op}: the shape conditions give the category as before, and the predicate attaches a weight to each

morphism, subject to

$$\gamma(u, \mathrm{id}_u) \;\sqsubseteq\; e, \qquad \gamma(u,\, x \cdot y) \;\sqsubseteq\; \gamma(u, x) \otimes \gamma(m_u x,\, y),$$

writing $m_u x$ for the codomain of the morphism x out of u. Both inequalities run this way because the counit and the comultiplication are morphisms out of the comonoid and the Dialectica condition lets the predicate only increase along a morphism, whereas enrichment asks for the two reverse inequalities, and that is why the base is the opposite lineale. This is the monomial case of the theorem of [8, Thm. 4.3], whose enrichment is over an opposite for the same reason. At the trivial lineale $L = 1$ the predicate carries nothing and both conditions hold vacuously, so what comes back is the statement of Ahman and Uustalu for monomials.

Read in the other direction, with morphisms satisfying $\beta(fu, y) \sqsubseteq \alpha(u, F(u, y))$, the counit and the comultiplication give the two reverse inequalities,

$$e \sqsubseteq \gamma(u, \mathrm{id}_u), \qquad \gamma(u, x) \otimes \gamma(m_u x, y) \sqsubseteq \gamma(u, x \cdot y),$$

and a comonoid for $\lhd$ is a small category enriched in L itself. Over the Lawvere quantale $([0, \infty], \geq, +, 0)$, where $\sqsubseteq$ is the numerical $\geq$, these say that identities weigh 0 and that a composite weighs no more than its parts together, so the weights are lengths satisfying the triangle inequality along composable morphisms. The structures of Proposition 11 and the duoidal structure of Theorem 13 do not depend on the reading, since every structure map there preserves the predicate with equality; the closure of $\otimes$ is the exception, since in the reversed order it would ask for $\otimes$ to be residuated.

The categories arising here have the same set of morphisms out of every object, since the challenges of a dialectica object do not vary with the witness, so what is caught is the monoids and the categories of constant out-degree rather than all small categories. Recovering the unrestricted statement asks for the challenges to depend on the witness, which is to leave the monomials.

The correspondence extends to the morphisms, but not to the ones a reader expects. A map of $\lhd$-comonoids goes forward on objects and, at each object, backward on the morphisms out of its image, so it is not a functor but a cofunctor, also called a retrofunctor. The backward direction is the one a Dialectica morphism has as well, which is why the two notions sit together as comfortably as they do, and on the enriched side the maps are enriched cofunctors.

For $\dagger A$ the category is a simple one. Its objects are the witnesses of A, the morphisms out of u are the words of challenges to u, composition

is concatenation, and the comultiplication sends every word back to the same object, so there are no morphisms between distinct witnesses. Read as a transition system, with the objects as states and the morphisms out of a state as the moves available there, it is a state that no sequence of challenges moves away from, which is the reading of † as storage that survives repeated use; the enrichment records how well the witness withstands the sequence. For $!A$ the same holds with the word replaced by a multiset, so the record of what has been used is kept but its order is not.

6 Order and sequentiality

Section 5 left the two modalities side by side and unrelated. What should relate them is the statement that whatever may be reused in parallel may be reused in sequence: $!A$, which is a comonoid for $\otimes$, ought to be a comonoid for $\triangleleft$ as well. Nothing in the definitions of the two tensors gives that, and a comonoid structure does not transport from one tensor to another without a map between them. The map is ser, which sends (u, v) to (u, const_v) and so makes the continuation ignore the challenge it is handed.

The substitution connective also has an older logical motivation. In the Linear Logic Model of State (LLMS) of [6, 7], contexts are built with two operations: a comma, which leaves its two parts unordered, and a semicolon, which puts everything in the first part before everything in the second. A structural rule, Ser, allows a context to be given more order while keeping the same formula occurrences. Thus the comma and semicolon generate the series-parallel orders carried by LLMS contexts.

Categorically, the simplest instance of this rule is

$$\mathsf{ser}A, B \colon A \otimes B \longrightarrow A \triangleleft B,$$

which adds an order between two previously unordered components. More generally it gives the middle interchange

$$\mathsf{int}A, B, C, D \colon (A \triangleleft B) \otimes (C \triangleleft D) \longrightarrow (A \otimes C) \triangleleft (B \otimes D),$$

which resequences two ordered contexts running in parallel.

Bechet, de Groote and Retoré [2] show that the inclusion of one series-parallel order in another is completely axiomatised and, once the two constructors share a unit and the comma is commutative and associative, generated by the middle interchange alone. The orders in question are the least class containing the point and closed under disjoint union and ordinal sum,

which is what the comma and the semicolon of LLMS build. So the map to ask of a model is int rather than ser, since ser and the two weak distributivities are instances of it, the first with B and C taken to be I, and one natural transformation is the whole of what the order costs. Both maps are already carried by the Dialectica construction.

Proposition 12. *There are morphisms in* $\mathbf{Dial}_L(\mathbf{Set})$*, natural in all arguments,*

$$\mathsf{ser}_{A,B} \colon A \otimes B \longrightarrow A \triangleleft B, \qquad \mathsf{int}_{A,B,C,D} \colon (A \triangleleft B) \otimes (C \triangleleft D) \longrightarrow (A \otimes C) \triangleleft (B \otimes D).$$

Proof. For ser, send the witness (u, v) to (u, const_v) and a challenge (x, y) to itself; the two predicates are then literally the same element $\alpha(u, x) \otimes \beta(v, y)$ of L. For int, send the witness $\big((u, \varphi), (w, \psi)\big)$ to $\big((u, w),\ (x, z) \mapsto (\varphi x, \psi z)\big)$ and a challenge $((x, z), (y, t))$ to $((x, y), (z, t))$; both predicates evaluate to $\alpha(u, x) \otimes \beta(\varphi x, y) \otimes \gamma(w, z) \otimes \delta(\psi z, t)$, up to the commutativity of $\otimes$ in L. For naturality, let morphisms be given in each argument. Both composites around the square have the same map on witnesses, since in either order a witness is relabelled by the given forward maps and then paired or curried; and the same map on challenges, since in either order a challenge is permuted and then carried back by the given backward maps. Two morphisms of $\mathbf{Dial}_L(\mathbf{Set})$ with the same two components are equal. □

Theorem 13. $\big(\mathbf{Dial}_L(\mathbf{Set}), I, \otimes, \triangleleft\big)$ *is a normal duoidal category: a symmetric monoidal structure* $(\otimes, I)$ *and a second monoidal structure* $(\triangleleft, I)$ *on the same unit, related by the monoidal natural transformation* int*.*

Proof. Theorem 3 gives the first tensor with its unit and its closure, Proposition 8 gives the second with the same unit, and Proposition 12 gives the interchange map. Four coherence conditions remain, two of associativity and two of units. The first says that the two routes from $((A \triangleleft B) \otimes (C \triangleleft D)) \otimes (E \triangleleft F)$ to $(A \otimes C \otimes E) \triangleleft (B \otimes D \otimes F)$, one applying int to the left pair first and the other to the right pair first, agree once the associativity of $\otimes$ is inserted. The second says the same for $\triangleleft$: the two routes from $((A \triangleleft B) \triangleleft C) \otimes ((A' \triangleleft B') \triangleleft C')$ to $((A \otimes A') \triangleleft (B \otimes B')) \triangleleft (C \otimes C')$ agree once the associativity of $\triangleleft$ is inserted. The two unit conditions are the cases of int in which a whole row or a whole column is the unit, and each asserts that the resulting map is the one the unit isomorphisms already give. Normality is the statement that the two comparison maps between the unit and itself, $I \otimes I \to I$ and $I \to I \triangleleft I$, are isomorphisms, and both are identities here: all three objects are $(1, 1, e)$, since $e \otimes e = e$.

Each of the four is an equality between two morphisms of $\mathbf{Dial}_L(\mathbf{Set})$. On witnesses and on challenges both sides are built by pairing, projection and currying out of the same data, and so agree as maps of sets; the predicate at corresponding arguments is on both sides the same $\otimes$-product of the same values of L, up to the associativity and commutativity of $\otimes$. Since a morphism here is determined by its two components, the two sides are equal in each case. Taking B and C to be I in int, and using the unit isomorphisms, gives ser. □

Corollary 14. *Every comonoid for $\otimes$ in $\mathbf{Dial}_L(\mathbf{Set})$ is a comonoid for $\triangleleft$, with the same counit and with comultiplication $C \to C \otimes C \xrightarrow{\mathsf{ser}} C \triangleleft C$. In particular $!A$ is a comonoid for $\triangleleft$, its comultiplication sending u to (u, const_u) and adding multisets backwards.*

Proof. The counit is available because the two tensors share the unit I, so the counit $C \to I$ of the $\otimes$-comonoid serves unchanged. For coassociativity, both ways round the pentagon send a witness of C to the same triple, since ser leaves the witness of each component untouched and only converts a pair into a rule ignoring its challenge, and carry a challenge back by the same composite, since ser is the identity on challenges; the predicate conditions were already checked for the $\otimes$-comonoid and for ser. For $!A$ the composite is the map displayed in the statement. □

This discharges the debt of Section 5, and it is what an LLMS-category has to require of a model by hand: there the transfer is imposed, by asking for a monic ser that makes $!A$ a comonoid for "before", and here it comes with the axioms.

7 Conclusions

A Dialectica object has a shape and a predicate, and the comparison with polynomial functors separates them. The shape records the witnesses, the challenges and the ways one may depend on the other; the predicate records which witness withstands which challenge. The functor $|-|\colon \mathbf{Dial}(\mathbf{Set}) \to \mathbf{Poly}_{\mathsf{mono}}$ keeps the shape and forgets the predicate, and it is faithful without being full, which is the precise sense in which the polynomial is what remains visible of a proposition.

All the interesting connectives that **Poly** carries on monomials were already Dialectica connectives more than 30 years ago. They come from one piece of work, the thesis [10], and from the two papers that preceded

it [4, 9]: two constructions, two categories, each monoidal closed in its own right, each connective given with its predicate as part of a model of linear logic. Table 1 is the summary of the comparison. Read across a row and one sees a connective, its witnesses and its challenges, and then the name the polynomial literature has for the shape that is left when the predicate is dropped.

Substitution is the row that looks like an exception, but it is not one. The operation is old on both sides: it is composition of polynomials, and on the Dialectica side it was within reach from the start. It was set aside because the tensors were being kept commutative, and nothing in a model of intuitionistic linear logic asked for a tensor that was not. What asks for one is a calculus with ordered contexts, and that is what gives the connective its reading here. A witness of $A \triangleleft B$ answers a challenge to A with the witness of B to be confronted next, and the two predicates are combined in the order the play imposes. The connective was implied by the constructions before it was written down, since the passage from no dependence $X \times Y$ to dependence on both sides $X^V \times Y^U$ already ran through them in the challenge coordinate, and negation puts the same passage in the witness coordinate. Once $\triangleleft$ is there, $\dagger$ is its comonoid and the passage from reuse in parallel to reuse in sequence is a consequence rather than an assumption, showcasing old work of Bechet-de Groote-Retoré as well.

What none of the polynomial vocabulary can say is which witness withstands which challenge, and that is what makes a proposition out of a pattern of interaction. The lesson is not that Dialectica constructions can be replaced by polynomial ones. It is that the polynomial exposes one component of a structure Dialectica has had from the beginning, and exposing it makes the patterns of dependence easier to recognise and to organise. What it cannot do is tell two propositions apart when they share an interaction pattern and differ in their predicates.

Two companion notes take up what is left out here. One asks what a lens obliged to respect a predicate can do that a plain lens cannot, and is written for the bidirectional-transformations community. The other asks what has to be required of a monad on the challenge coordinate, and which functional interpretations the answer then covers.

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